

See also: The United Nations and International Court of Justice 212–219

■ The Endangered Species Act 264–265 ■ The Kyoto Protocol 305



species diversity is crucial to sustain our planet's web of life, the WNBR now aims to show how biodiversity and human cultural diversity are mutually beneficial.

Environmental sustainability rests on the premise that if people have an economic or cultural stake in their environment—for example, they rely on harvesting its fish for food or its timber for building—they are more likely to protect it and ensure that stocks are not depleted. In many BRs, local people also benefit from ecotourism.

Building the network

A global network is important to truly reflect Earth's biodiversity. Governments can nominate new BRs. If MAB's governing body agrees that they meet the required criteria, they are added to the WNBR. Each BR is protected by the environmental laws of the country in which it is situated.

Yellowstone National Park, in the states of Wyoming, Montana, and Idaho, was designated a Biosphere Reserve in 1976.

In the 50 years since the MAB program was launched, the threats facing the biosphere have intensified. One million species face extinction in the next few decades, including 40 percent of amphibians and 30 percent of marine mammals. Human-induced climate change will exacerbate the decline, which some scientists describe as Earth's sixth mass extinction.

In 2020, there were 701 BRs in 124 countries, ranging in size from the enormous Central Amazon BR in Brazil to Biosfer Dyfi BR, a small estuary in south Wales. Globally coordinated and legally enforceable measures such as the WNBR program offer the best way of meeting the huge environmental challenges facing the planet. ■

Russia's zapovedniks

Russia was a world leader in creating wildlife sanctuaries. Its first state-organized *zapovednik* (Russian for nature reserve) was set up in Barguzinsky, near Lake Baikal, in 1916, to conserve its population of sables—small mammals valued for their fur. In 2020, there were more than 100 *zapovedniks*, covering about 1.4 percent of the country's total area. Some are vast: the Great Arctic Zapovednik, for example, covers 4 million ha (almost 10 million acres) of tundra and supports polar bears, snowy owls, seals, whales, and more.

Russian soil scientist Vasily Dokuchaev first put forward the idea for *zapovedniks* in the 1890s, proposing that people other than scientists should be excluded from them. Russian botanist Ivan Borodin later argued that they should not be established piecemeal, but should be planned to include every main ecosystem. These reserves provide research environments where scientists can see nature unaffected by human influence.

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We should preserve every scrap of biodiversity as priceless while we learn ... what it means to humanity.

E. O. Wilson

American biologist, 1992

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